

# CMS Open Data H to tau tau Search: Final Analysis Note

Analysis my\_analysis

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## Abstract

This note documents a reduced CMS 2012 Open Data search for Higgs boson decays to tau pairs in the mu tau\_h final state. A Phase 5 audit found that the original observed BDT-score fit was not a valid evidence result: it omitted a reducible QCD/fake-tau background estimate and failed high-score shape validation. The final conservative result therefore uses visible mass in VBF, boosted, and zero-jet categories with a same-sign data-driven QCD/fake template. It gives  $\mu_{\text{hat}} = 0.4382$ , an observed 95% CLs limit  $\mu < 10.7645$ , and  $Z = 0.0949$ . The categorized BDT-score model is retained as a flagged diagnostic with  $\mu_{\text{hat}} = 9.3518$  and  $Z = 3.9416$ .

## Change Log

Phase 5 v2 responds to the observed-versus-expected audit. It adds a same-sign QCD/fake estimate, changes the primary full-data result to the visible-mass fallback, keeps the BDT score as a diagnostic, adds NN/MVA performance figures, and compiles the paper with REVTeX PRL formatting.

## 1 Data And Simulation

The data are Run2012B and Run2012C TauPlusX reduced samples from the public HiggsTauTauReduced mirror, using the CMS Open Data H to tau tau luminosity reference of 11.467/fb (CMS Collaboration 2021; CMS Open Data Analyses 2021). The localized mirror files contain about one tenth of the event entries listed in the Open Data

records; this is treated as the public reduced processing sample scope, while MC weights continue to use the official record denominators specified in `normalization_inputs.json`, as required by the sample provenance.

The available MC set contains ggH and VBF H to tau tau, DYJetsToLL, TTbar, and W1/W2/W3JetsToLNu. QCD multijet, diboson, single top, embedded Z to tau tau, W4/inclusive W+jets, associated Higgs, and H to WW remain unavailable as reduced samples and are not silently substituted.

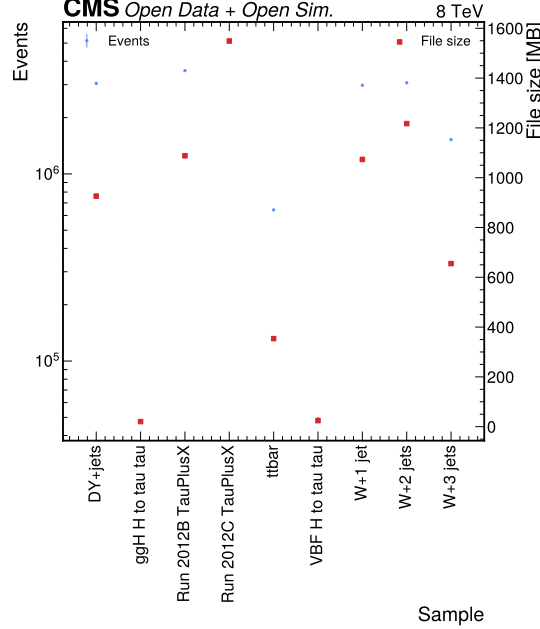


Figure 1: Sample inventory. The figure summarizes local reduced ROOT entries and file sizes for the public H to tau tau files used in the analysis. The entries are processing entries, while normalization uses official Open Data record denominators and the cited luminosity reference.

## 2 Event Selection And Categories

Events are required to pass `HLT_IsoMu17_eta2p1_LooseIsoPFTau20`. Muons have  $pt > 20$  GeV and  $|\eta| < 2.1$  with tight identification and isolation. Hadronic tau candidates have  $pt > 20$  GeV,  $|\eta| < 2.3$ , decay-mode ID, tight isolation ID, and tight anti-muon ID. The signal region uses opposite-sign mu tau<sub>h</sub> pairs with muon-MET transverse mass below 40 GeV. The W control region uses high transverse mass, and the QCD/fake estimate uses same-sign low-mT events.

The fit categories are mutually exclusive VBF, boosted, and zero-jet. VBF requires at least two clean jets, leading-dijet mass above 300 GeV, and leading-dijet  $|\Delta\eta|$  above 2.5. Boosted events have at least one clean jet after failing VBF, and zero-jet events have no clean jets.

| Category | Data | Background | QCD/fake | Signal | Data/background |
|----------|------|------------|----------|--------|-----------------|
| vbf      | 78   | 96.28      | 17.75    | 1.590  | 0.810           |
| boosted  | 2183 | 2364.39    | 325.58   | 9.170  | 0.923           |
| zero_jet | 8451 | 8456.81    | 1569.65  | 14.341 | 0.999           |

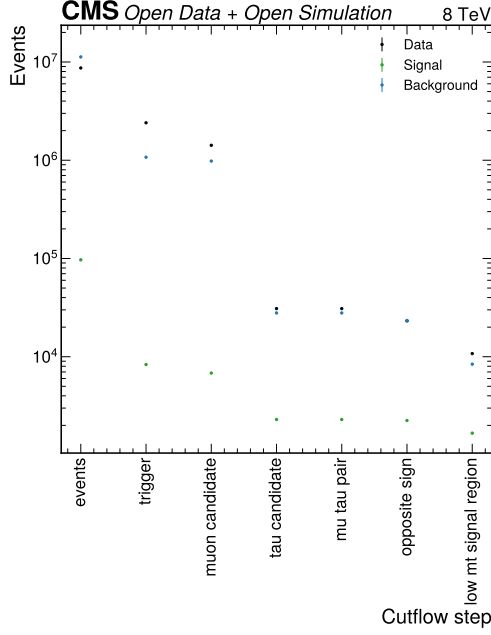


Figure 2: Cutflow summary. This plot shows raw event counts after each selection step for data, signal MC, and background MC. It verifies that the selection chain is monotonic and that the reduced samples remain populated after trigger, object, charge, and transverse-mass requirements.

### 3 QCD And W Control Corrections

The W+jets normalization is measured in the high-mT control region as  $0.8528 \pm 0.0370$ . The 10% validation value was  $0.8433 \pm 0.0868$ . The values agree within the larger 10% uncertainty and support the W-control arithmetic.

The VBF category originally had a large MC overprediction. Tightening the VBF cuts and applying a simple b-tag veto did not solve it, so a disjoint VBF-like top-btag control region outside the low-mT signal region was used to calibrate the MC-background rate in the VBF category. The resulting VBF background scale is  $0.5571 \pm 0.0515$ ; it is applied only to MC backgrounds in VBF and not to the signal or QCD/fake template.

The same-sign QCD/fake estimate subtracts non-QCD MC from same-sign low-mT data and transfers the resulting template to opposite-sign signal candidates. For the primary visible-mass model, the OS/SS transfer factor measured in the lowest visible-mass bin is  $0.9958 \pm 0.1201$ . For the BDT score diagnostic, the corresponding low-score factor is  $0.8173 \pm 0.0853$ .

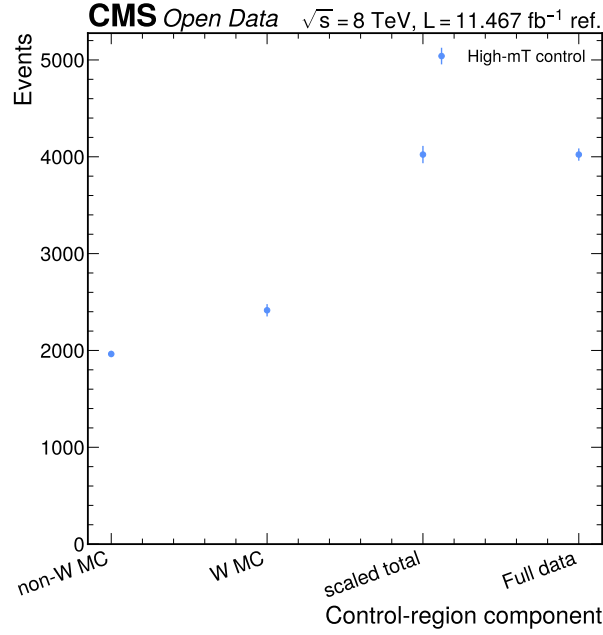


Figure 3: Full high-mT W control comparison. The figure shows the derivation inputs for the full-data W+jets control scale. The control region is outside the low-mT signal region and the scale is propagated to the observed workspace without signal-region tuning.

## 4 MVA And Transformer Diagnostics

The requested NN/MVA program was evaluated using the available reduced event features. Histogram gradient boosting, an MLP neural network, and XGBoost were trained on selected MC only. A transformer was not trained because the current pixi environment lacks a fast attention stack and the reduced files do not contain the GenMET target needed for the requested missing-momentum regression.

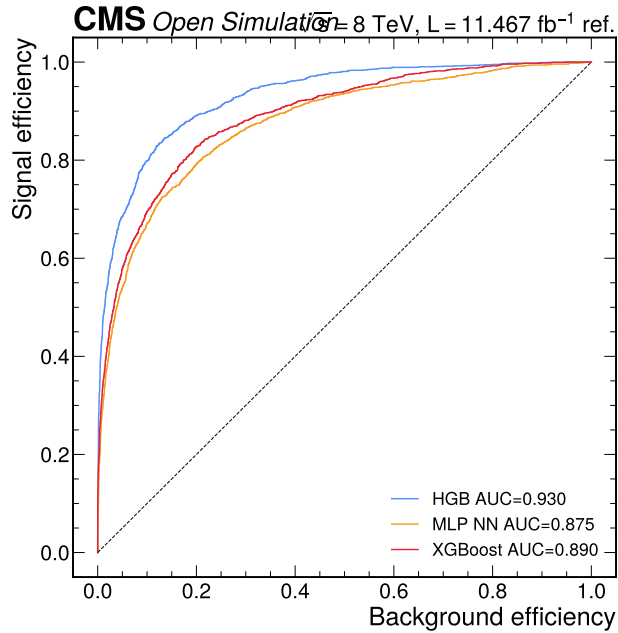


Figure 4: MVA ROC curves. This figure shows ROC curves for the HGB, MLP neural-network, and XGBoost score columns on selected signal/background MC. It is an open-simulation performance diagnostic and does not validate data/MC score shapes.

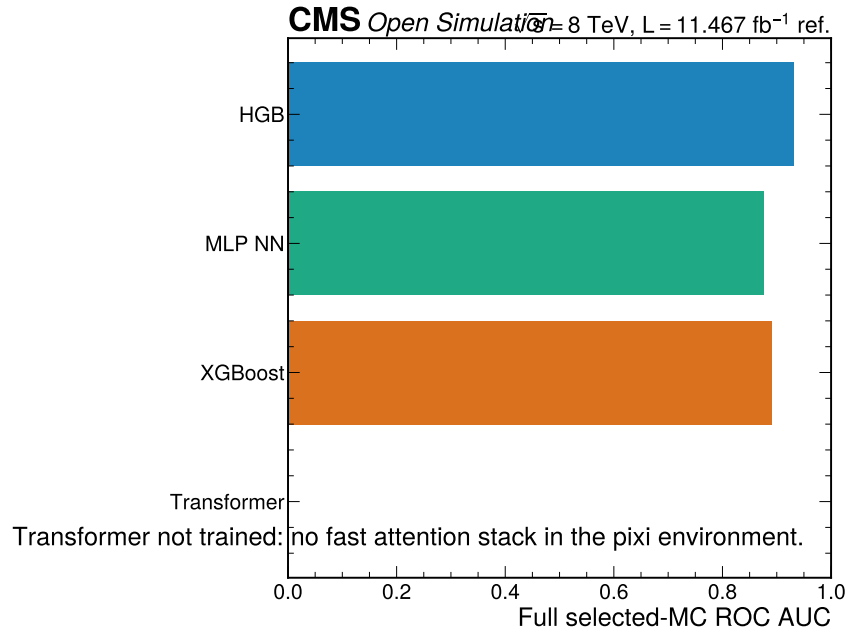


Figure 5: MVA AUC summary. This figure compares the full selected-MC ROC AUC values for the trained classifiers and records that the transformer branch was not trained in this fast reduced-sample pass. The BDT score has strong MC separation, but that is not sufficient for an observed evidence claim.

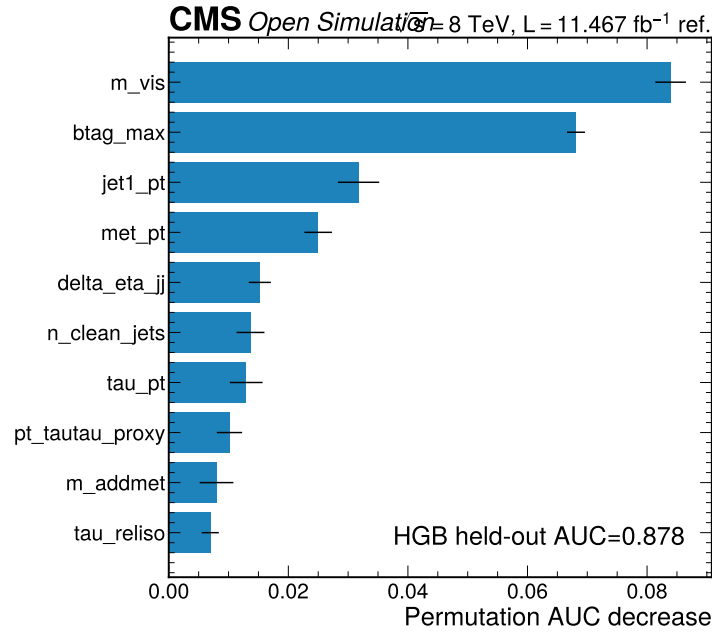


Figure 6: HGB permutation importance. This figure shows the leading permutation importance values for a retrained HGB classifier on a held-out selected-MC split. It documents which reduced features drive the classifier response used in the diagnostic score fit.

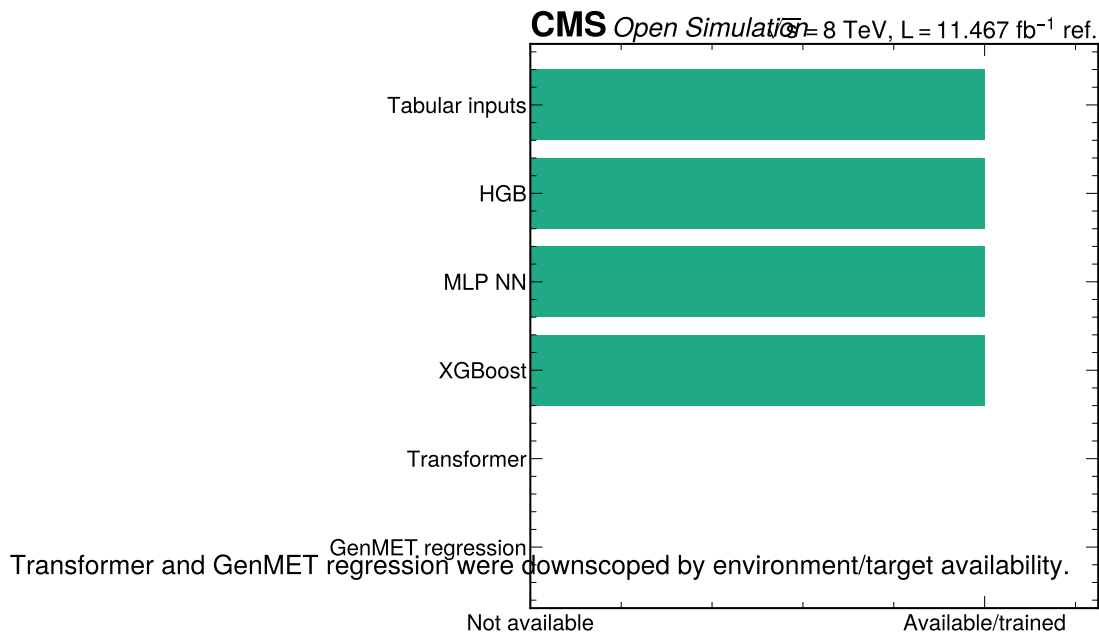


Figure 7: Transformer feasibility. This figure records which modern-model branches were available in the current environment. The transformer and GenMET-regression branches are explicitly downscoped rather than reported as unperformed successes.

## 5 Primary Visible-Mass Result

The primary statistical model is a binned pyhf profile likelihood in visible mass, with one signal-strength parameter  $\mu$ , the  $W$  high-mT scale, the same-sign QCD/fake transfer uncertainty, luminosity, DY and tau/open-data normalization terms, and MC statistical terms (Heinrich et al. 2021; Read 2002; Cowan et al. 2011).

| Quantity                              | Value           | Interpretation                         |
|---------------------------------------|-----------------|----------------------------------------|
| Primary median expected 95% CLs limit | $\mu < 10.7375$ | corrected visible-mass workspace       |
| Primary observed 95% CLs limit        | $\mu < 10.7645$ | conservative observed result           |
| Primary $\mu_{\text{hat}}$            | 0.4382          | best fit, bounded at zero if no excess |
| Primary $q_0 Z$                       | 0.0949          | diagnostic only                        |
| Primary combined data/background      | 0.9812          | validation after QCD/fake correction   |
| Primary $\chi^2/\text{ndf}$           | 0.7224          | validation after QCD/fake correction   |

The primary result is much more stable than the original score-only observed fit. It does not show a Higgs-like excess:  $\mu_{\text{hat}}$  is 0.4382 and  $Z = 0.0949$ . The observed limit is compatible with the weak sensitivity expected from this reduced single-final state setup.

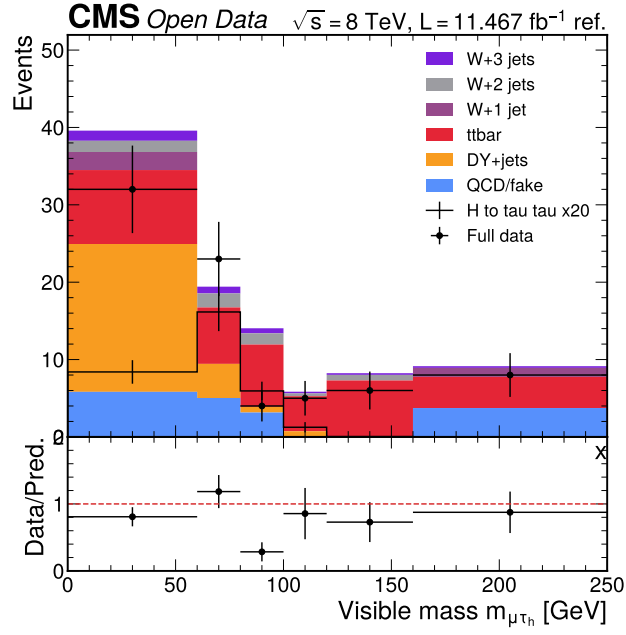


Figure 8: Primary visible-mass validation in VBF. The plot compares full data to the QCD-corrected visible-mass prediction in the VBF category. The remaining VBF deficit is a limitation, but it no longer drives a fake signal-strength increase.

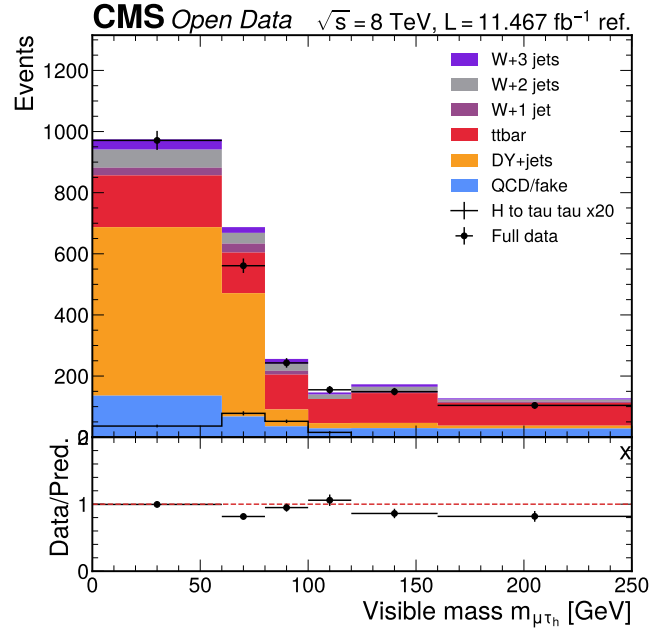


Figure 9: Primary visible-mass validation in boosted. The plot compares full data to the QCD-corrected visible-mass prediction in the boosted category. This channel is part of the conservative primary fit.

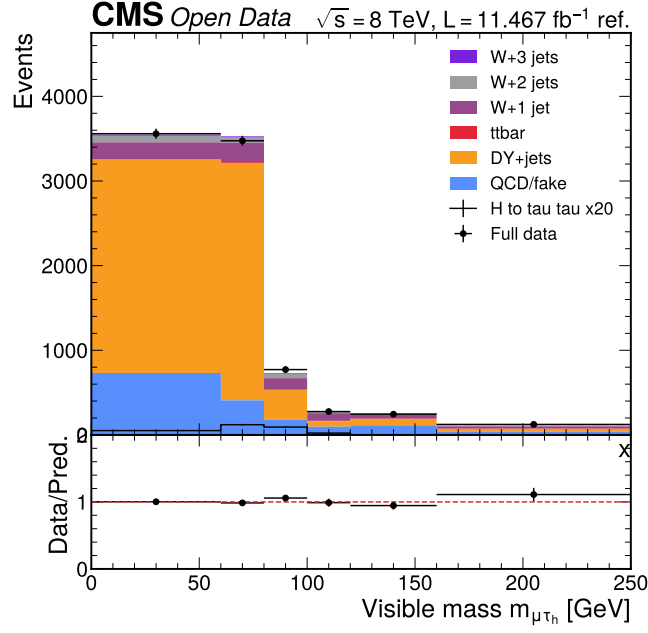


Figure 10: Primary visible-mass validation in zero-jet. The zero-jet normalization is stabilized by the same-sign QCD/fake estimate, which removes the dominant normalization pathology seen in the original score-only result.

## 6 BDT Score Diagnostic

The BDT-score model is kept because it was explicitly requested and has better expected MC separation. It is not used as the primary observed evidence result, because its high-score bins remain shape-flagged after adding the QCD/fake template.

| Category | Data | Background | QCD/fake | Signal | Data/background |
|----------|------|------------|----------|--------|-----------------|
| vbf      | 79   | 94.14      | 12.43    | 1.590  | 0.839           |
| boosted  | 2213 | 2331.71    | 268.90   | 9.175  | 0.949           |
| zero_jet | 8466 | 8187.20    | 1292.40  | 14.341 | 1.034           |

| Quantity                            | Value           | Interpretation                             |
|-------------------------------------|-----------------|--------------------------------------------|
| Score median expected 95% CLs limit | $\mu < 4.1802$  | diagnostic expected score workspace        |
| Score observed 95% CLs limit        | $\mu < 14.8373$ | flagged diagnostic                         |
| Score $\mu_{\text{hat}}$            | 9.3518          | flagged high-score shape diagnostic        |
| Score $q_0 Z$                       | 3.9416          | not CMS-quality evidence                   |
| Score combined data/background      | 1.0137          | score validation after QCD/fake correction |
| Score $\chi^2/\text{ndf}$           | 1.1750          | score validation after QCD/fake correction |

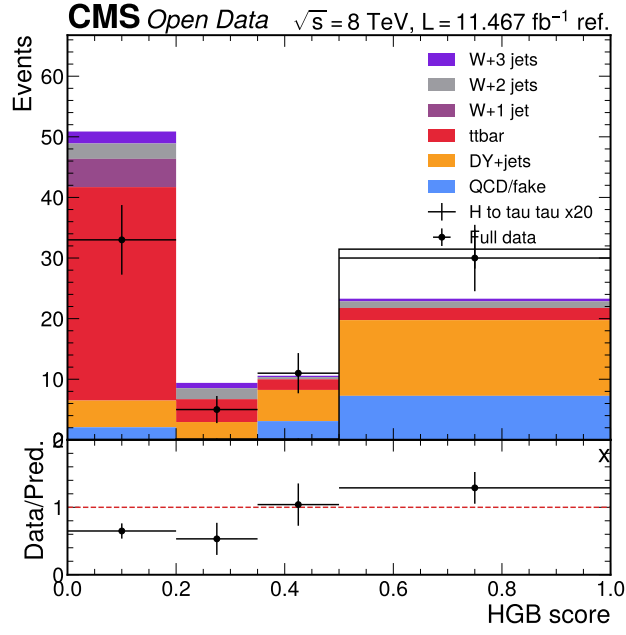


Figure 11: Score-template diagnostic in VBF. This figure keeps the categorized BDT score view requested for sensitivity studies. It is diagnostic because the score shape is not validated well enough for an observed evidence claim.

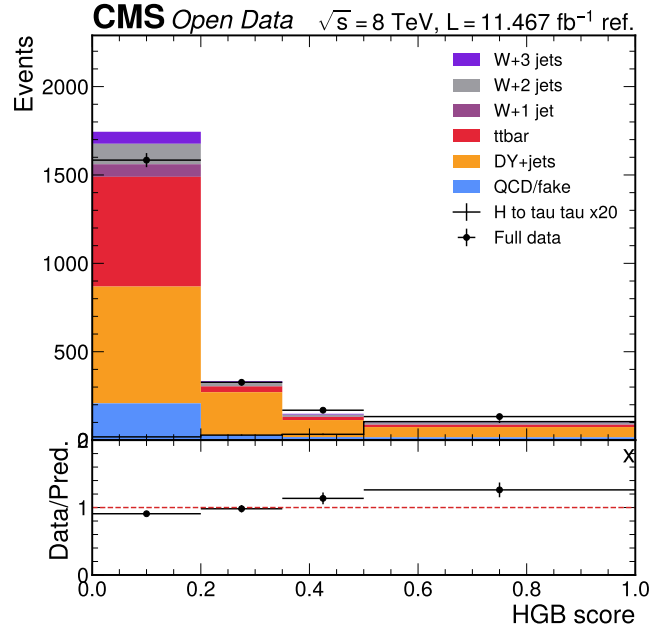


Figure 12: Score-template diagnostic in boosted. This figure shows the boosted score template after QCD/fake correction. High-score residuals motivate the diagnostic-only status.

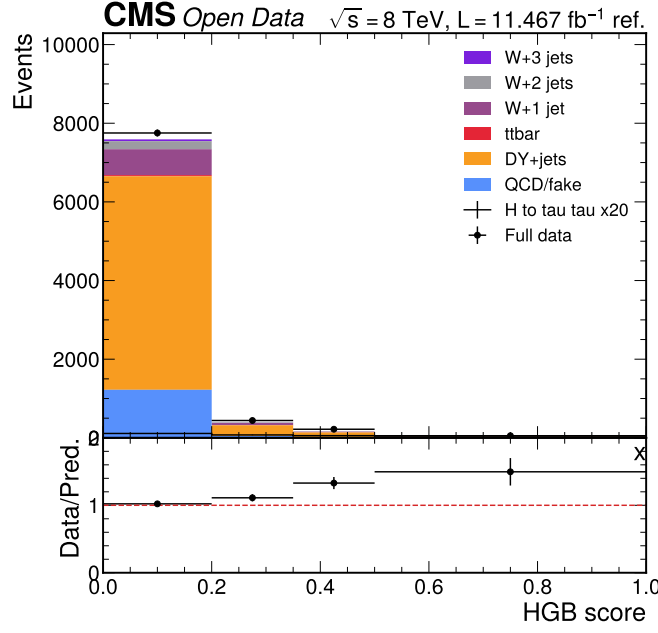


Figure 13: Score-template diagnostic in zero-jet. This figure shows that the zero-jet normalization is improved by QCD/fake correction while high-score shape residuals remain visible.

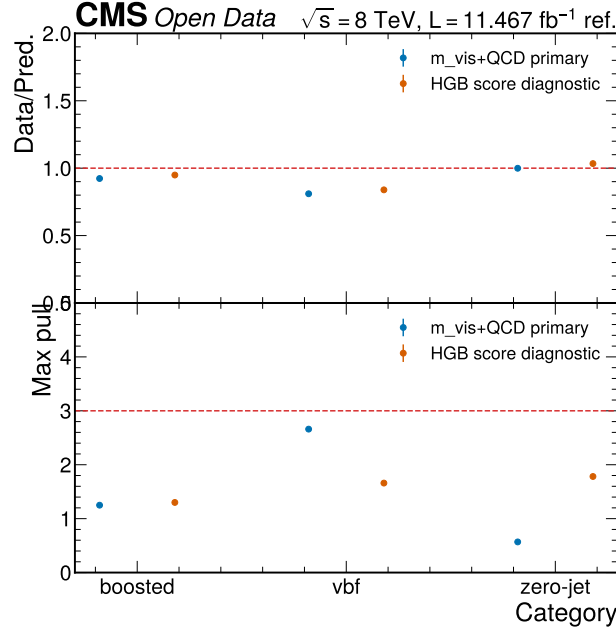


Figure 14: Observed pull and ratio summary. The figure compares primary visible-mass and BDT-score validation behavior after the QCD/fake audit correction. It shows why the visible-mass model is primary and the score fit is diagnostic.

## 7 Comparison With Published Results

CMS Run 1 reported evidence for H to tau tau using 7 and 8 TeV data with multiple final states, embedded backgrounds, and a full calibration program; the quoted best-fit signal strength was  $0.78 \pm 0.27$  (CMS Collaboration 2014). CMS 2018 reported observed and expected significances of 4.9 and 4.7 and a signal strength near 1.09 (CMS Collaboration 2018). These are not direct pass/fail targets for this reduced open-data workflow.

| Result                 | Scope                                       | Significance               | Signal-strength information                                           |
|------------------------|---------------------------------------------|----------------------------|-----------------------------------------------------------------------|
| This open-data primary | 8 TeV mu tau_h reduced mirror, visible mass | 0.095                      | $\mu_{\text{hat}} = 0.438$ ; $\mu < 10.764$ ; expected $\mu < 10.738$ |
| This BDT diagnostic    | 8 TeV mu tau_h reduced mirror, HGB score    | 3.942 flagged              | $\mu_{\text{hat}} = 9.352$ ; $\mu < 14.837$ ; expected $\mu < 4.180$  |
| CMS Run 1              | 7+8 TeV multi-channel                       | $>3$                       | $\mu = 0.78 \pm 0.27$                                                 |
| CMS 2018               | 13 TeV multi-channel                        | observed 4.9, expected 4.7 | $\mu = 1.09 \pm \text{about } 0.27$                                   |

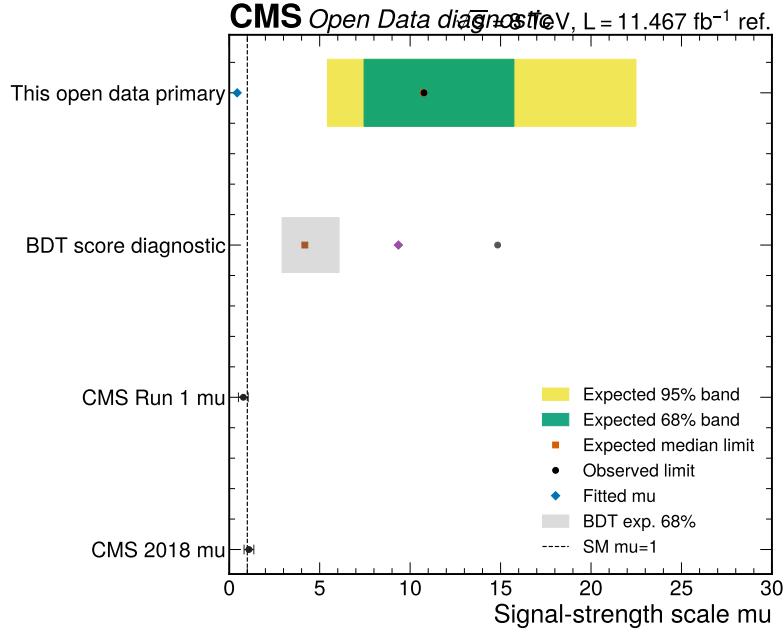


Figure 15: Signal-strength and limit comparison. The figure uses a CMS-style expected-band and observed-marker presentation for the primary open-data limit, with the flagged BDT diagnostic and published CMS signal-strength measurements shown as context. The open-data diagnostic is not directly equivalent to CMS publication measurements.

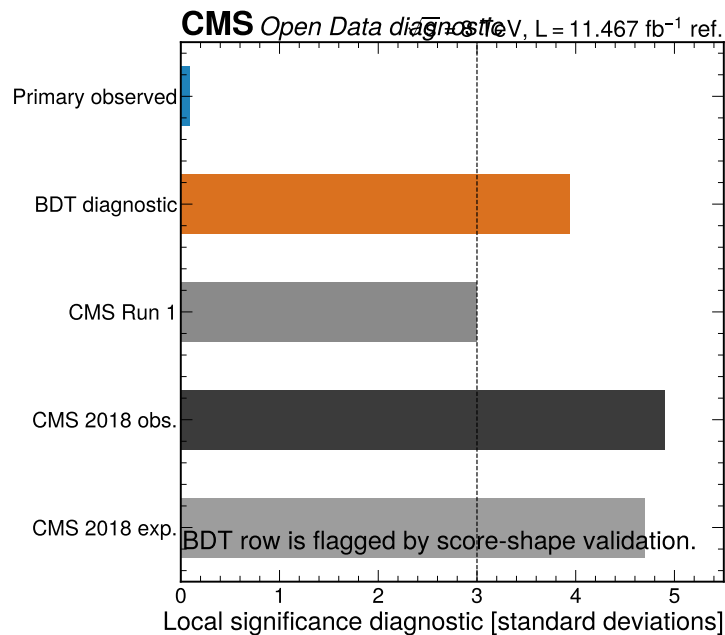


Figure 16: Significance comparison. The figure compares the primary open-data diagnostic and the flagged BDT-score diagnostic with CMS publication values. The BDT row is explicitly not interpreted as evidence.

## 8 Conclusion

The final audit-corrected result is a conservative visible-mass plus QCD/fake template fit. It gives  $\mu_{\text{hat}} = 0.4382$ , an observed 95% CLs limit  $\mu < 10.7645$ , and  $Z = 0.0949$ . The originally surprising BDT-score observed result is retained only as a flagged diagnostic because the score shape is not validated in data. This is the correct interpretation of the reduced CMS Open Data workflow: reproducible and useful for methodology, but not CMS-quality evidence for H to tau tau.

## References

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